CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Lax epimorphisms and the associated factorization

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Abstract

Lax epimorphisms (also called co-fully-faithful morphisms) are a 2-dimensional version of epimorphisms; in a 2-category A they are precisely the 1-cells f making A(f,C) fully faithful for all objects C. In this talk, several features of lax epimorphisms will be presented. We show that any 2-category with convenient colimits has an orthogonal LaxEpi-factorization system and give a concrete description of this factorization in Cat. In the more general context of V-enriched categories, we give several characterizations of lax epimorphisms in V-Cat. This is joint work with Fernando Lucatelli Nunes.